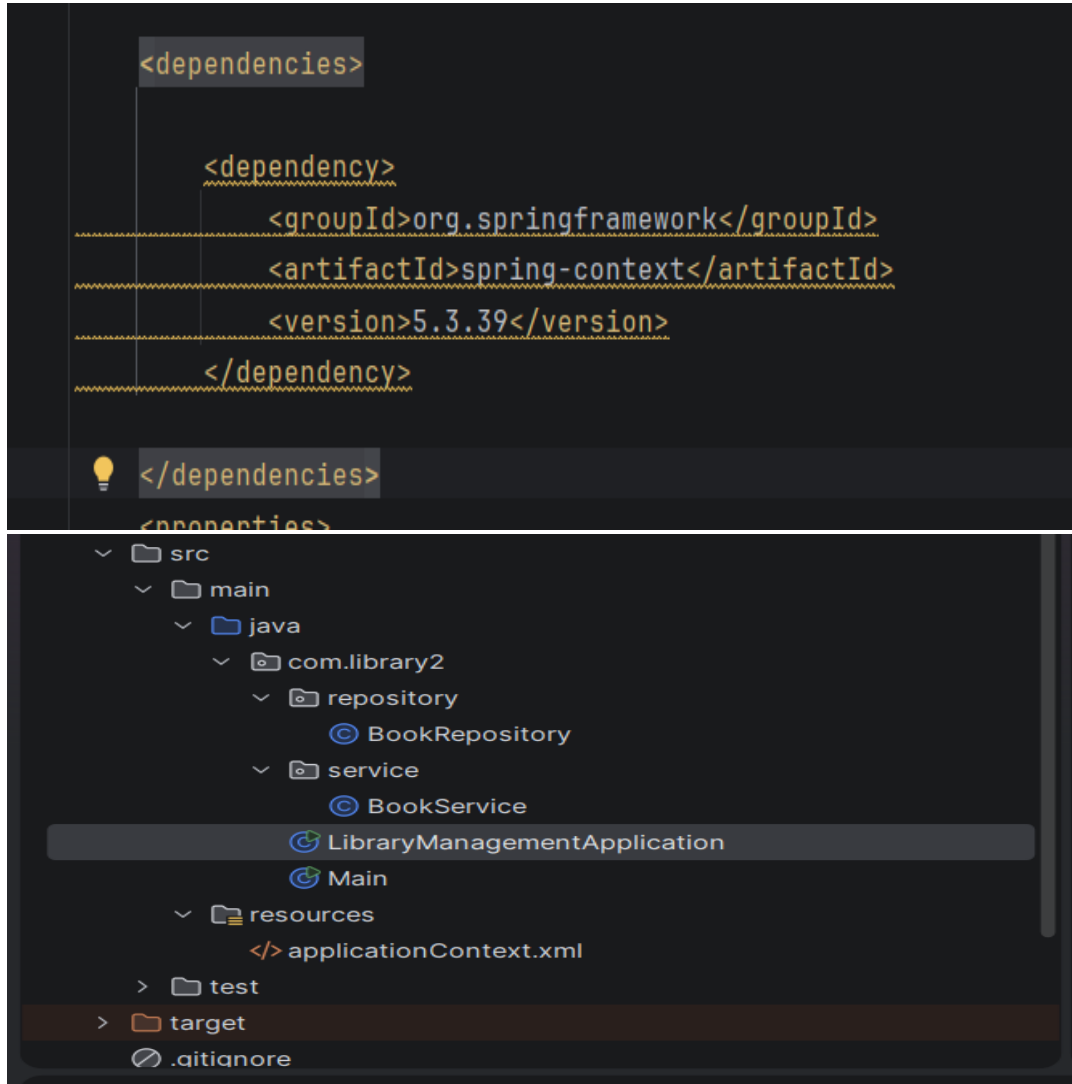


Week 3

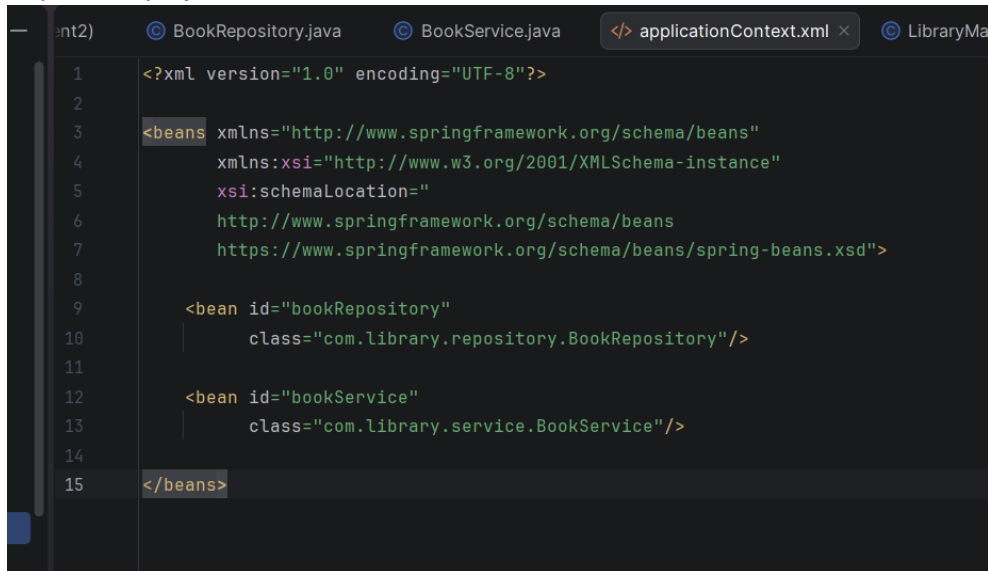
LibraryManagementSystem Project

Exercise 1

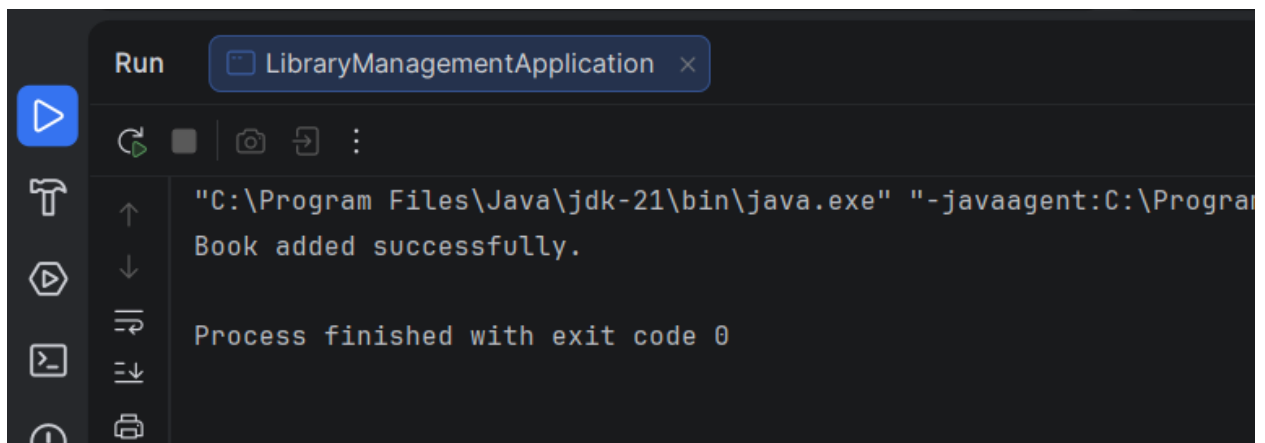
adding Spring core dependencies



Dependency injection

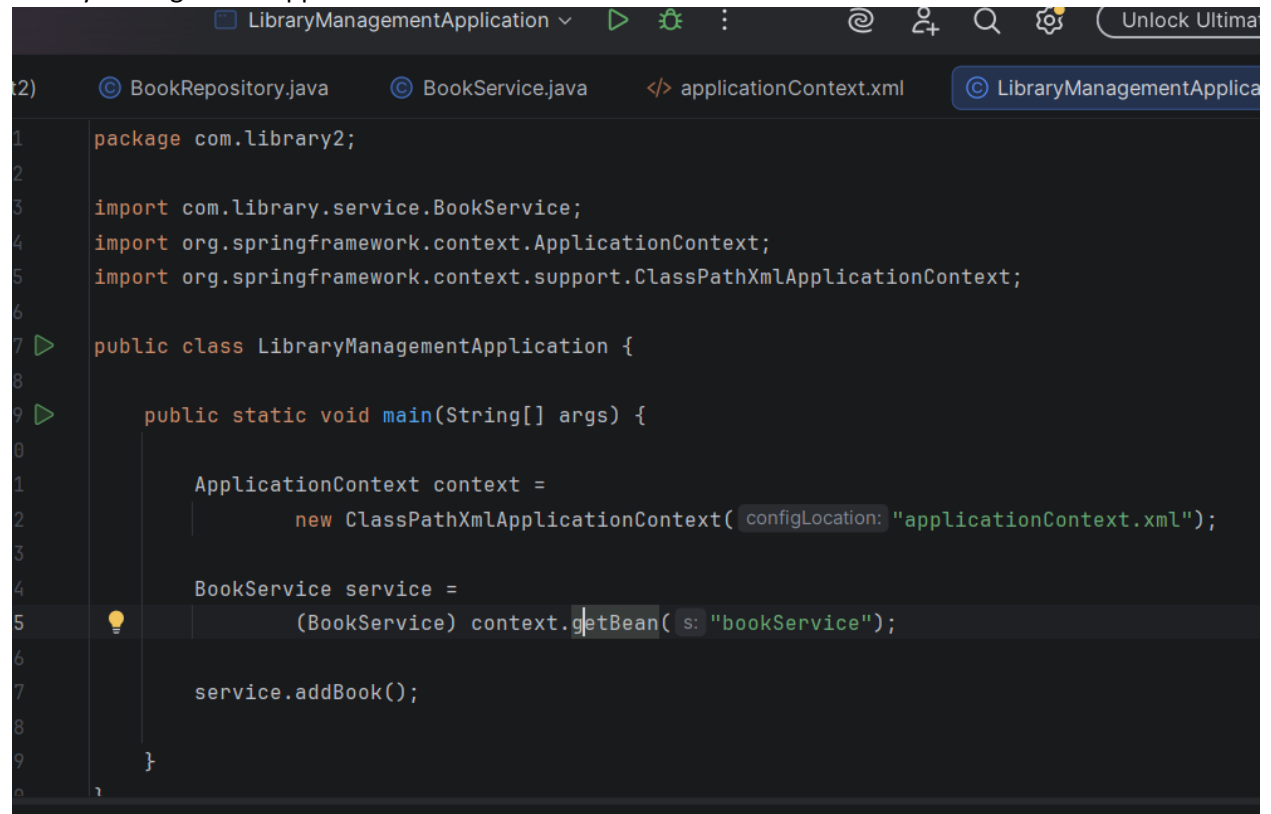


```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="bookRepository"
10         class="com.library.repository.BookRepository"/>
11
12     <bean id="bookService"
13         class="com.library.service.BookService"/>
14
15 </beans>
```



```
Run LibraryManagementApplication x
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\Java\jdk-21\bin\javaagent.jar"
Book added successfully.
Process finished with exit code 0
```

Library Management Application main code



The screenshot shows an IDE window with the title bar "LibraryManagementApplication". The editor displays the main code of the application, which is a Java class named `LibraryManagementApplication` in the package `com.library2`. The code imports `com.library.service.BookService`, `org.springframework.context.ApplicationContext`, and `org.springframework.context.support.ClassPathXmlApplicationContext`. The `main` method creates an `ApplicationContext` instance, retrieves a `BookService` bean, and calls `addBook()` on it. A lightbulb icon is visible next to the `getBean` call, indicating a suggestion or warning.

```
1 package com.library2;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class LibraryManagementApplication {
8
9     public static void main(String[] args) {
10
11         ApplicationContext context =
12             new ClassPathXmlApplicationContext("applicationContext.xml");
13
14         BookService service =
15             (BookService) context.getBean("bookService");
16
17         service.addBook();
18
19     }
20 }
```